RAJALAKSHMI ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

2026-2027- Odd Semester

III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

FraudShield AI- Intelligent Digital Payment Risk Assessment System

First Review

BarathRaj M (241901015) | Deepan S A (241901021)
Supervisor: Dr. S. Bhuvana, Assistant Professor [SG]

Introduction

FraudShield AI is an intelligent digital payment risk assessment system designed to identify potentially fraudulent transactions.

It analyzes transaction parameters such as payment amount, transaction time, device trust, geographic location, login behavior, transaction frequency and account activity.

The system uses Knowledge Representation, Predicate Logic and Forward Chaining to reason about suspicious transaction patterns.

The system classifies transactions into Low, Medium, High and Critical risk levels and provides explainable reasons with appropriate security recommendations.

Problem Statement

Digital payments are increasingly exposed to fraudulent transactions.Existing systems may not effectively combine multiple transaction risk factors.There is a need for an intelligent system to identify suspicious transaction patterns.The system should provide logical, explainable and risk-based fraud assessment.

Motivation

Selected to address the increasing risk of fraud in digital payment transactions.
The project combines cybersecurity with Artificial Intelligence concepts such as Knowledge Representation and rule-based reasoning.
The system aims to identify suspicious transaction patterns and provide transparent risk assessment.
Expected benefits include improved fraud identification, explainable decisions and appropriate security actions.

Objectives

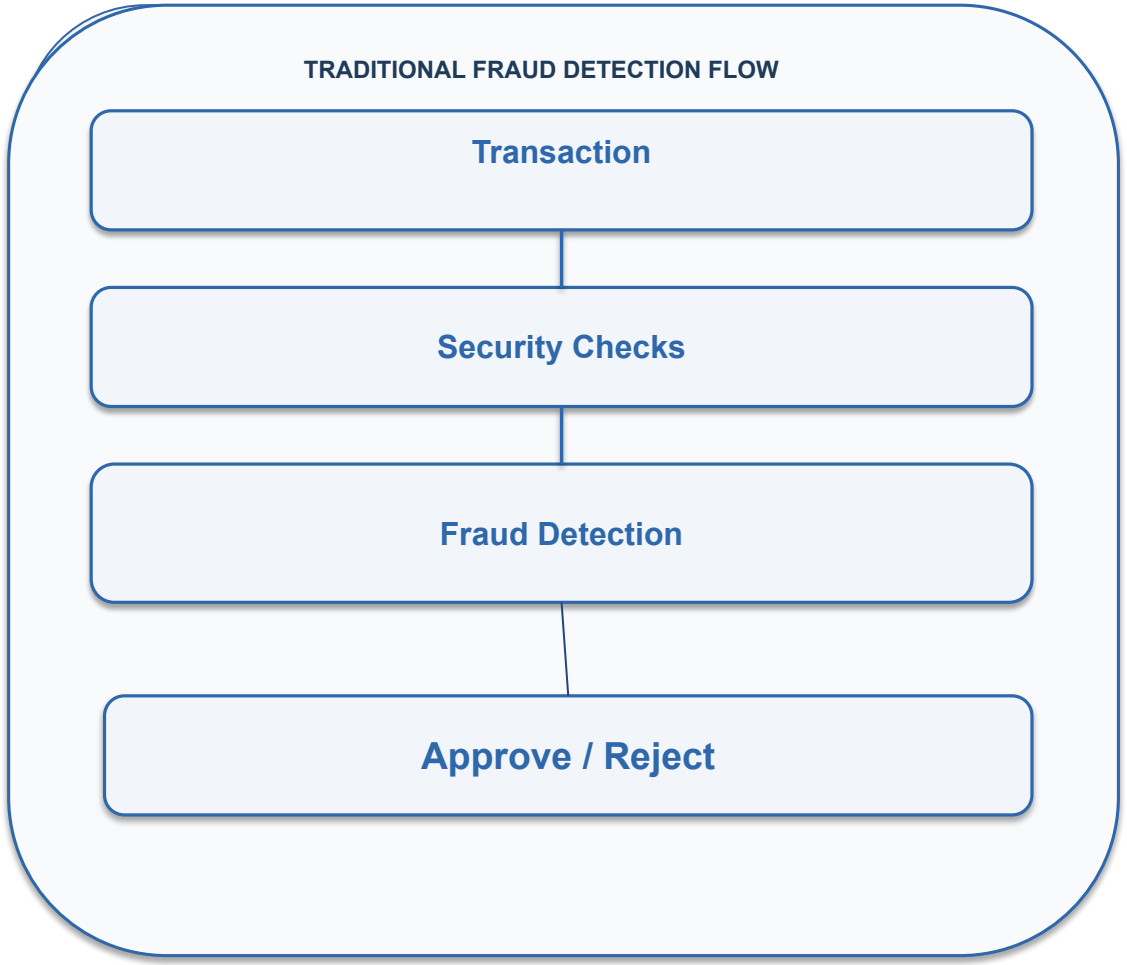
- To develop an AI-based digital payment fraud risk assessment system.
- To analyze multiple transaction and behavioral parameters.
- To represent transaction information using Predicate Logic.
- To apply Forward Chaining for rule-based risk inference.
- To classify transactions into Low, Medium, High and Critical risk levels.
- To provide explainable decisions and recommended security actions.

Literature Survey

S.No.	Author & Year	Method / Technique	Key Findings	Limitations
1	Giarratano & Riley	Expert Systems & Rule-Based Reasoning	Explains knowledge bases, rules and inference mechanisms for expert systems.	Requires domain-specific rules and knowledge.
2	Arrieta et al. (2020)	Explainable AI	Emphasizes transparency and interpretability of AI decisions.	General XAI approach rather than payment-specific reasoning.
3	Russell & Norvig (2021)	Knowledge Representation & Reasoning	Provides foundations for representing knowledge and reasoning using AI.	General AI framework; not specific to digital payment fraud.

Existing System

Digital payment platforms use automated fraud detection and security checks.
Typical flow: Transaction ⟶ Security Checks ⟶ Fraud Detection ⟶ Decision
Advantages: Fast transaction monitoring and automated fraud detection.
Limitations: Limited integration of multiple risk factors, dependence on predefined rules or thresholds, and limited explanation of the final decision.

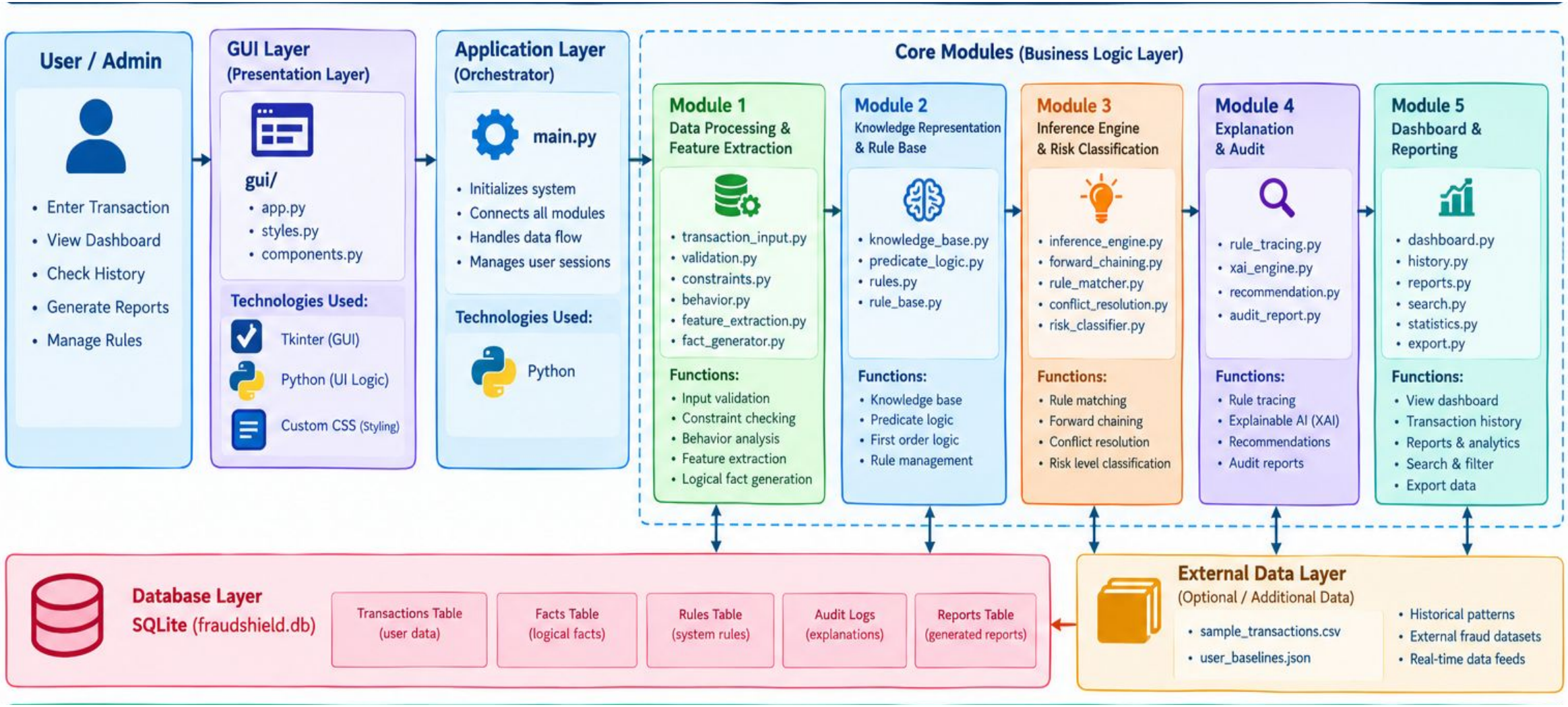


Proposed System

FraudShield AI analyzes multiple transaction and behavioral parameters for fraud risk assessment.
The system validates transaction data and converts it into structured facts.
Predicate Logic represents transaction knowledge and predefined fraud rules.
Forward Chaining applies the rules to derive the transaction risk level.
The system classifies transactions as Low, Medium, High or Critical and provides an explainable decision with a recommended security action.



System Architecture / Block Diagram



Methodology

1. Transaction Input & Data Collection

2. Parameter Validation & Constraint Checking

3. Feature Extraction & Fact Generation

4. Knowledge Representation using Predicate Logic

5. Rule Matching & Forward Chaining

6. Risk Classification & Explainable Decision

Technologies Used

Programming Language: Python 3.11+
AI Technique: Rule-Based Expert System, Forward Chaining
Knowledge Representation: Predicate Logic / First-Order Logic
Database: SQLite
Interface: Python GUI
Tools / IDE: Visual Studio Code, Git & GitHub
Testing: Python Unit Testing
Hardware: Intel Core i5 or above, 8 GB RAM, 256 GB SSD

Module Description

Module 1 – Transaction Data Acquisition & Parameter Validation

Module 2 – Knowledge Representation Engine

Module 3 – AI Gateway & Provider Management

Module 4 – Explainable AI & Decision Engine

Module 5 – Database, Dashboard & Reporting

Results

- Validate digital payment transaction parameters.
- Identify suspicious transaction patterns using multiple risk factors.
- Generate logical facts using Predicate Logic.
- Apply Forward Chaining for rule-based risk inference.
- Classify transactions into Low, Medium, High and Critical risk levels.
- Provide explainable decisions with recommended security actions.
- Store transaction data and analysis results for future reference.

Project Schedule

Activity	Month 1	Month 2	Month 3	Month 4
Literature Survey	✓			
Problem Identification	✓			
System Design	✓	✓		
Module Implementation		✓	✓	
Testing & Evaluation			✓	✓

Current Progress

- ✓ Literature survey completed
- ✓ Problem identification completed
- ✓ Requirement analysis and methodology finalized
- ✓ Proposed system architecture designed
- ✓ Five core modules defined
- ✓ Module 1 database structure designed
- ☐ Complete system implementation
- ☐ Testing and final evaluation

Future Work

- Complete the implementation of all five FraudShield AI modules.
- Implement Predicate Logic and Forward Chaining based inference.
- Develop risk classification and explainable decision mechanisms.
- Integrate database, dashboard and transaction history.
- Conduct testing using normal and suspicious transaction scenarios.
- Evaluate system performance and prepare final documentation.

Conclusion

FraudShield AI proposes an intelligent rule-based system for digital payment risk assessment.
The system combines transaction validation, Knowledge Representation, Predicate Logic and Forward Chaining.
It analyzes multiple transaction and behavioral parameters to identify suspicious patterns.
The proposed system provides risk classification along with explainable decisions and recommended security actions.

References

[1] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson, 2021.
[2] J. Giarratano and G. Riley, Expert Systems: Principles and Programming, 4th Edition, Cengage Learning.
[3] A. Barredo Arrieta et al., “Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI,” Information Fusion, vol. 58, pp. 82–115, 2020.